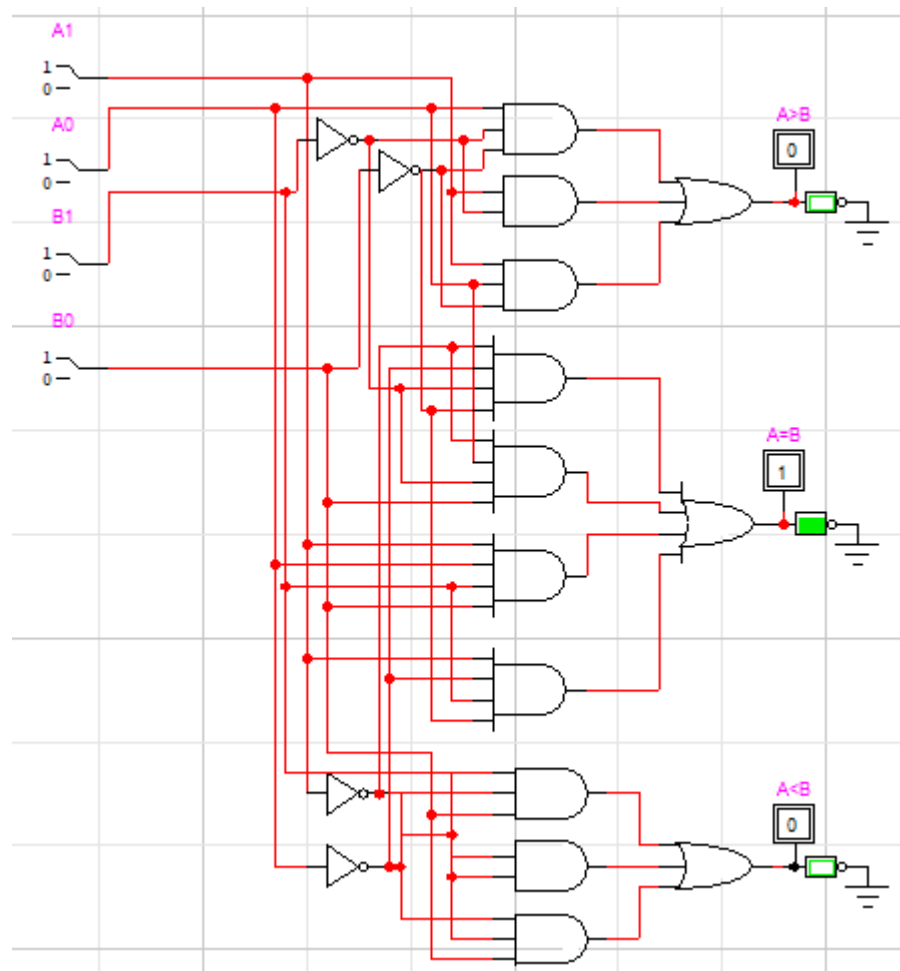


Task 1:

A_1	A_0	B_1	B_0	$A < B$	$A = B$	$A > B$
0	0	0	0	0	1	0
0	0	0	1	1	0	0
0	0	1	0	1	0	0
0	0	1	1	1	0	0
0	1	0	0	0	0	1
0	1	0	1	0	1	0
0	1	1	0	1	0	0
0	1	1	1	1	0	0
1	0	0	0	0	0	1
1	0	0	1	0	0	1
1	0	1	0	0	1	0
1	0	1	1	1	0	0

1	1	0	0	0	0	1
1	1	0	1	0	0	1
1	1	1	0	0	0	1
1	1	1	1	0	1	0

**Task 2:****Truth Table:**

Player A A, A, A ₀	Player B B, B, B ₀	Player A wins	Player B wins	Draw
000	000	0	0	1
000	001	0	1	0
000	010	0	1	0
000	011	0	1	0
000	100	0	1	0
000	101	0	1	0
000	110	0	1	0
000	111	0	1	0
001	000	1	0	0
001	001	0	0	1
001	010	0	1	0
001	011	0	1	0
001	100	0	1	0
001	101	0	1	0
001	110	0	1	0
001	111	0	1	0
010	000	1	0	0
010	001	1	0	0
010	010	0	0	1
010	011	0	1	0
010	100	0	1	0
010	101	0	1	0
010	110	0	1	0

010	111	0	1	0
011	000	1	0	0
011	001	1	0	0
011	010	1	0	0
011	011	0	0	1
011	100	0	1	0
011	101	0	1	0
011	110	0	1	0
011	111	0	1	0

K-Map:

Player B Wins

Date _____

A \ B	000	001	011	010	110	111	101	100	
000	1	1	1	1	1	1	1	1	$A_1'A_2'A_3 + A_1'A_2B_1 + A_1'B_2 + A_2'A_3 + A_2B_1B_3$
001	1	1	1	1	1	1	1	1	$A_2'A_3 + B_2B_1B_3$
011					1	1	1	1	
010			1		1	1	1	1	
110									
111									
101									
100									

Player A

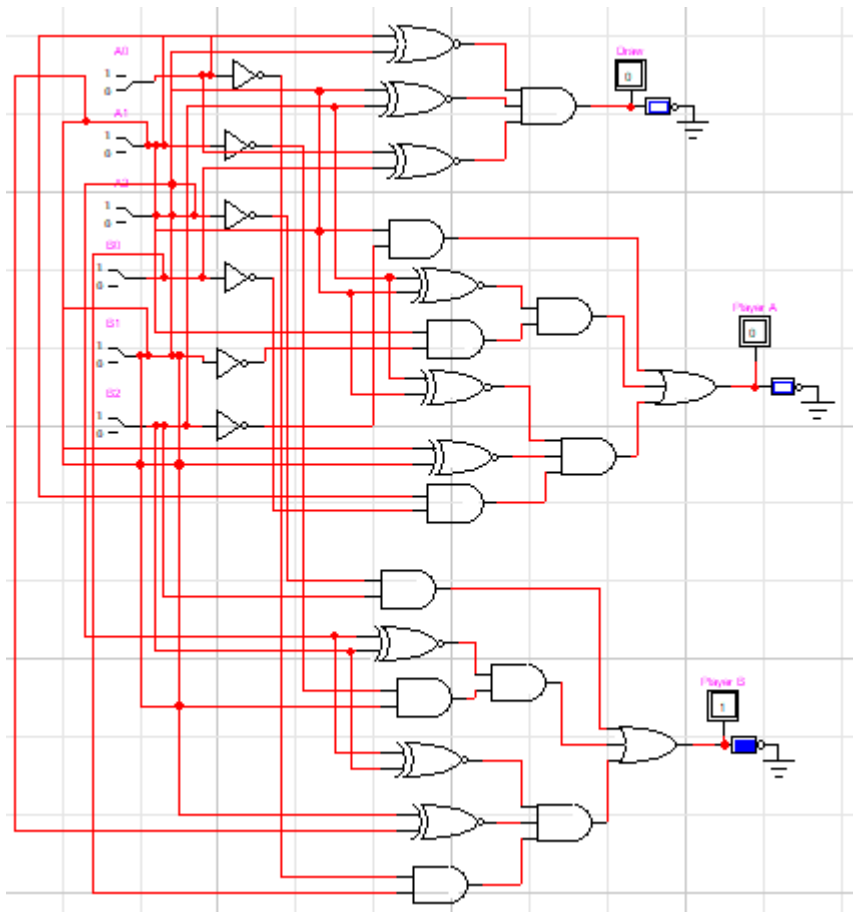
A \ B	000	001	011	010	110	111	101	100	
000									
001	1			0					
011	1	1		1					
010	1	1							
110									
111									
101									
100									

$$A_1'A_2B_1B_3 + A_1'A_2B_2B_1 + A_1'A_2A_3B_1B_3$$

DRAW

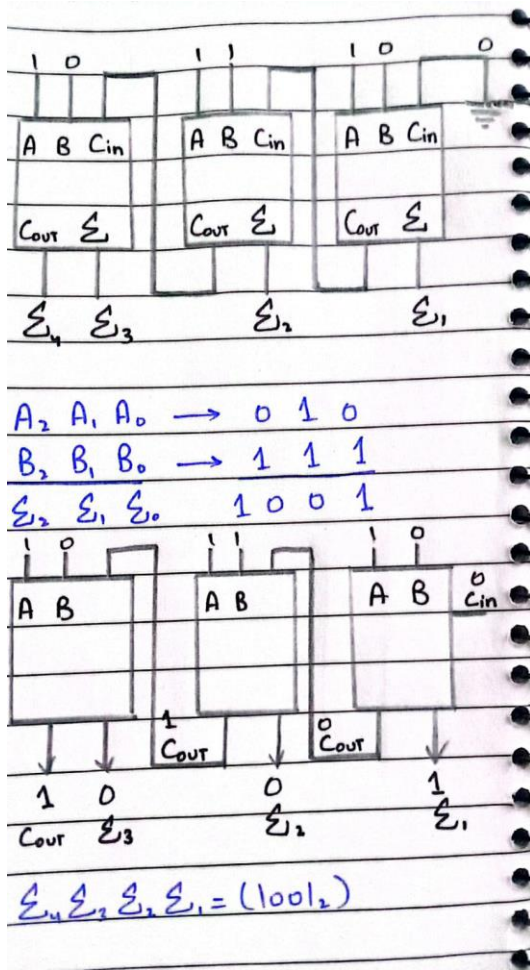
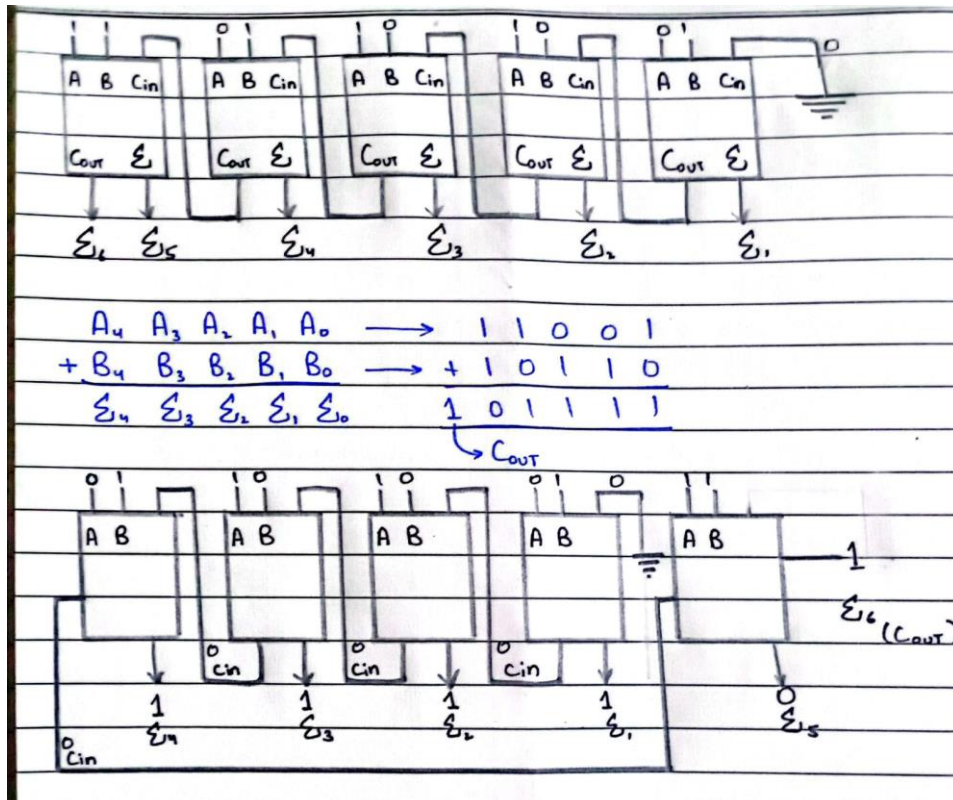
A \ B	000	001	011	010	110	111	101	100	
000	1								
001		1							
011			1						
010				1					
110									
111									
101									
100									

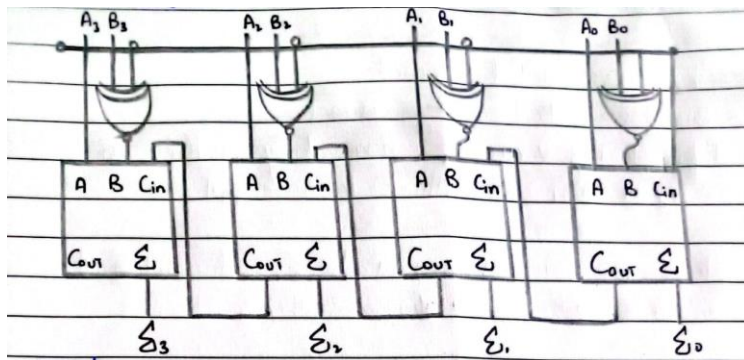
$$A_1'A_2A_3B_2B_1B_3 + A_1'A_2A_3B_2B_1B_3 + A_1'A_2A_3B_2B_1B_3 + A_1'A_2A_3B_2B_1B_3$$

**Task 3:**

Comp			
0	1		
1	1	A	A > B — 1
1	1		
0	0		
0	0		

Task 4:

**Task 5:****Task 6:**



$\overline{\text{Add}}/\text{Sub} = 1$

$A = 1010$

$B = 1101$

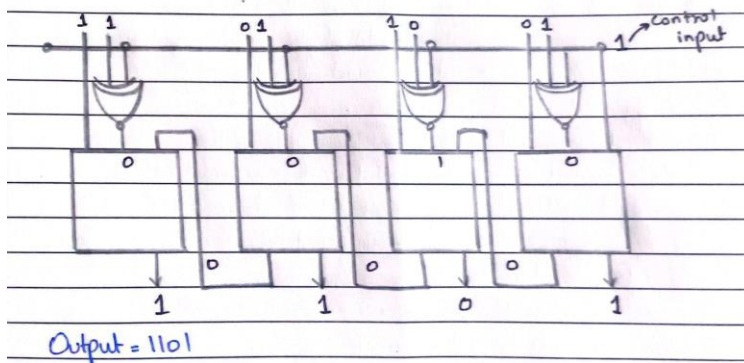
Whenever Control input is 1 its subtraction

2's of $B = 0010$

+ 1
0011

Now add = 1010

0011
1101



Output = 1101

Task 7:

$\overline{\text{Add}}/\text{Sub} = 0$

$A = 1010$

$B = 1101$

whenever Control input = 0 its addition

1 0 1 0

+ 1 1 0 1

OUTPUT = 1011

1 0 1 1
Cout

